

DIY Robotic Arm Precautions

Note: When the robotic arm grips an object, you need to control the gripper angle. Improper angle settings may cause the servo to stall and burn out.

Here we have compiled a gripper angle reference table, documenting the approximate servo angle required to grip objects of 0.5 cm increments.

Below we provide a table showing the approximate servo angles for gripping objects of different sizes.

Note: Due to servo mechanical structure tolerances, the data cannot be guaranteed to be completely accurate. Adjust according to your actual situation. If the servo stalls and overheats, immediately disconnect the power.



Object Length (Unit: cm)	Servo Angle (Unit: degrees)
0	180
0.5	176
1.0	168
1.5	160
2.0	152
2.5	143
3.0	134
3.5	125
4.0	115
4.5	105
5.0	95
5.5	80
6.0	57
6.0-6.4	0-57

For example, the vision recognition cube we provide has dimensions of 3cm in length, width, and height, so setting the servo angle to 134° is generally sufficient. Never set the angle excessively large.

